RV32I Reference Cards

Usage Template		Type	Description	Detailed Description
add	rd, rs1, rs2	R	Add	rd ← rs1 + rs2, pc ← pc+4
addi	rd, rs1, imm	I	Add Immediate	rd ← rs1 + imm_i, pc ← pc+4
and	rd, rs1, rs2	R	And	rd ← rs1 ∧ rs2, pc ← pc+4
andi	rd, rs1, imm	I	And Immediate	rd ← rs1 ∧ imm_i, pc ← pc+4
auipc	rd, imm	U	Add Upper Immediate to PC	rd ← pc + imm_u, pc ← pc+4
beq	rs1, rs2, pcrel_13	B	Branch Equal	pc ← pc + ((rs1==rs2) ? imm_b : 4)
bge	rs1, rs2, pcrel_13	B	Branch Greater or Equal	pc ← pc + ((rs1>=rs2) ? imm_b : 4)
bgeu	rs1, rs2, pcrel_13	B	Branch Greater or Equal Unsigned	pc ← pc + ((rs1>=rs2) ? imm_b : 4)
blt	rs1, rs2, pcrel_13	B	Branch Less Than	pc ← pc + ((rs1<rs2) ? imm_b : 4)
bltu	rs1, rs2, pcrel_13	B	Branch Less Than Unsigned	pc ← pc + ((rs1<rs2) ? imm_b : 4)
bne	rs1, rs2, pcrel_13	B	Branch Not Equal	pc ← pc + ((rs1!=rs2) ? imm_b : 4)
csrrw	rd, csr, rs1	I	Atomic Read/Write	rd ← csr, csr ← rs1, pc ← pc+4
csrrs	rd, csr, rs1	I	Atomic Read and Set	rd ← csr, csr ← csr ∨ rs1, pc ← pc+4
csrrc	rd, csr, rs1	I	Atomic Read and Clear	rd ← csr, csr ← csr ∧ ∼rs1, pc ← pc+4
csrrwi	rd, csr, zimm	I	Atomic Read/Write Immediate	rd ← csr, csr ← zimm, pc ← pc+4
csrrsi	rd, csr, zimm	I	Atomic Read and Set Immediate	rd ← csr, csr ← csr ∨ zimm, pc ← pc+4
csrrci	rd, csr, zimm	I	Atomic Read and Clear Immediate	rd ← csr, csr ← csr ∧ ∼zimm, pc ← pc+4
ecall		I	Environment Call	Transfer Control to Debugger
ebreak		I	Environment Break	Transfer Control to Operating System
jal	rd, pcrel_21	J	Jump And Link	rd ← pc+4, pc ← pc+imm_j
jalr	rd, imm(rs1)	I	Jump And Link Register	rd ← pc+4, pc ← (rs1+imm_i) & ∼1
lb	rd, imm(rs1)	I	Load Byte	rd ← sx(m8(rs1+imm_i)), pc ← pc+4
lbu	rd, imm(rs1)	I	Load Byte Unsigned	rd ← zx(m8(rs1+imm_i)), pc ← pc+4
lh	rd, imm(rs1)	I	Load Halfword	rd ← sx(m16(rs1+imm_i)), pc ← pc+4
lhu	rd, imm(rs1)	I	Load Halfword Unsigned	rd ← zx(m16(rs1+imm_i)), pc ← pc+4
lui	rd, imm	U	Load Upper Immediate	rd ← imm_u, pc ← pc+4
lw	rd, imm(rs1)	I	Load Word	rd ← sx(m32(rs1+imm_i)), pc ← pc+4
or	rd, rs1, rs2	R	Or	rd ← rs1 ∨ rs2, pc ← pc+4
ori	rd, rs1, imm	I	Or Immediate	rd ← rs1 ∨ imm_i, pc ← pc+4
sb	rs2, imm(rs1)	S	Store Byte	m8(rs1+imm_s) ← rs2[7:0], pc ← pc+4
sh	rs2, imm(rs1)	S	Store Halfword	m16(rs1+imm_s) ← rs2[15:0], pc ← pc+4
sll	rd, rs1, rs2	R	Shift Left Logical	rd ← rs1 << (rs2%XLEN), pc ← pc+4
slli	rd, rs1, shamt	I	Shift Left Logical Immediate	rd ← rs1 << shamt_i, pc ← pc+4
slt	rd, rs1, rs2	R	Set Less Than	rd ← (rs1 < rs2) ? 1 : 0, pc ← pc+4
slti	rd, rs1, imm	I	Set Less Than Immediate	rd ← (rs1 < imm_i) ? 1 : 0, pc ← pc+4
sltiu	rd, rs1, imm	I	Set Less Than Immediate Unsigned	rd ← (rs1 < imm_i) ? 1 : 0, pc ← pc+4
sltu	rd, rs1, rs2	R	Set Less Than Unsigned	rd ← (rs1 < rs2) ? 1 : 0, pc ← pc+4
sra	rd, rs1, rs2	R	Shift Right Arithmetic	rd ← rs1 >> (rs2%XLEN), pc ← pc+4
srai	rd, rs1, shamt	I	Shift Right Arithmetic Immediate	rd ← rs1 >> shamt_i, pc ← pc+4
srl	rd, rs1, rs2	R	Shift Right Logical	rd ← rs1 >> (rs2%XLEN), pc ← pc+4
srli	rd, rs1, shamt	I	Shift Right Logical Immediate	rd ← rs1 >> shamt_i, pc ← pc+4
sub	rd, rs1, rs2	R	Subtract	rd ← rs1 - rs2, pc ← pc+4
sw	rs2, imm(rs1)	S	Store Word	m32(rs1+imm_s) ← rs2[31:0], pc ← pc+4
xor	rd, rs1, rs2	R	Exclusive Or	rd ← rs1 ⊕ rs2, pc ← pc+4
xori	rd, rs1, imm	I	Exclusive Or Immediate	rd ← rs1 ⊕ imm_i, pc ← pc+4